The PLAMB Conjecture as a Test Programme for Redshift, Light Propagation, and Galaxy-Scale Acceleration

through hierarchical raw-MU offsets, redshift-frame preregistration, the full SPARC filtered-family posterior, and the 18 July FR charge-conjugation, environmental-replication and power-calibration audits

Prepared for the PLAMB research programme

18 July 2026 FR charge, environment and power revision

Research status

This paper treats PLAMB as a controversial working conjecture and an empirical test programme.

The numerical results reported here are recent internal diagnostics from July 12-16, 2026, combined with peer-reviewed and public datasets.

They should not be read as proof that special relativity is violated, that cosmic expansion is false, or that the Lambda-CDM model has been replaced.

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Abstract

The PLAMB conjecture is an exploratory proposal associated with Peter Lamb that asks whether part of the observed redshift-distance relation could arise from light-propagation, clock-rate, or inertia-projection effects rather than only from metric expansion and dark energy.

This paper summarises the conjecture as a falsifiable research programme, sets it beside known superluminal-looking phenomena in modern physics, and reports recent diagnostics against supernova, baryon acoustic oscillation, cosmic microwave background, quasar, gravitational-wave, and galaxy-rotation datasets.

The strongest recent positive results are:

(i) DES-Dovekie raw distance-modulus tests favour Peter Lamb's simple time-projection correction near beta = 0.5 with H0 around 67.3-68.0 km/s/Mpc under full covariance;

(ii) a July 14 joint DES-Dovekie, Pantheon+, and Union3.1 likelihood prefers dataset-level magnitude offsets by BIC and returns beta = 0.542 in the HD frame and beta = 0.527 in the HEL frame with H0 fixed at 67.5; a promoted hierarchical version with explicit zero-mean calibration priors finds that moderate dataset-plus-survey priors keep the HD H0-fixed beta-free fit near beta = 0.521, but the strongest rows remain calibration-mediated;

(iii) Union3.1 compressed distances remain a supportive but lower-resolution cross-check; and

(iv) a SPARC optical-depth branch connects the radial acceleration relation to an acceleration scale g0 = c H0 / 2pi and remains favoured over RAR in the Hubble-flow Q2 split after leverage removal.

July 14 hierarchy and predictive checks make that result more nuanced: the kappa,p branch fits p = 0.510, 0.522, and 0.483 across all-Q2, Hubble-flow Q2, and non-Hubble Q2, but conservative held-out scoring is near-tied outside Hubble-flow and low-acceleration outer points currently favour RAR.

The full filtered-family posterior, after removing six persistent negative high-leverage galaxies, keeps p near one half, p = 0.5193 [0.5144, 0.5237], and kappa = 0.7898 [0.7686, 0.8173]. A 16 July audit finds that the earlier negative stress scores were generated with prior-redrawn galaxy nuisance parameters rather than paired posterior draws, while robust rank-split diagnostics show that PLAMB is not converged.

DESI DR2 weakens earlier SU2/SU2R background evidence. Quaia now contains an object-level residual follow-up candidate after scan-law/colour regression, but the July 16 spatial-block null prevents promotion; GWTC-5.0 diagnostics do not yet support a decisive model promotion.

The filtered-family posterior therefore strengthens the p near one half pattern while weakening the case for a full-sample SPARC model promotion.

The resulting position is not a discovery claim. It is a research agenda: PLAMB is interesting because it now has sharp tests, clear positive cases, and clear failure modes.

The 18 July extension derives the complete FR charge-conjugation rule, finds the implemented SPARC dynamics charge-even, and therefore rejects an intrinsic matter/antimatter rotation-curve classifier. A preregistered local 2MRS environment association fails its predictive gate; an independently published group-catalogue predictor is null; and a 40,000-draw injection study shows only 36.50% empirically recalibrated joint power at the earlier |rho| = 0.307254 effect anchor.

Keywords: PLAMB conjecture; redshift systematics; supernova cosmology; SPARC; radial acceleration relation; DESI DR2; faster-than-light; no-signalling; variable speed of light; Hubble tension.

1. Introduction

Modern precision cosmology is built around a coherent framework: general relativity, homogeneous and isotropic expansion on large scales, cold dark matter, and a dark-energy component well described by Lambda-CDM in many datasets.

The framework is not merely a curve fit. It links supernova distances, baryon acoustic oscillations, the cosmic microwave background, large-scale structure, lensing, and local distance-ladder measurements.

Any alternative must therefore survive a wide network of cross-checks, not just one Hubble diagram.

PLAMB enters this landscape as a deliberately provocative hypothesis. In the current working form, it asks whether the observed distance and redshift relations include a projection caused by light propagation through changing conditions, effective clock-rate or inertia changes, or a Noether-like conservation rule involving G and c.

It has also grown a galaxy-scale branch in which an optical-depth variable can reproduce the empirical radial acceleration relation. The programme is controversial because it touches the interpretation of redshift, the meaning of c, and the distinction between coordinate effects and local causal limits.

This paper does not assume PLAMB is true. It does something more useful: it states the conjecture, compares it with known physics in which superluminal language is common but carefully limited, and summarises recent tests.

The goal is to make the hypothesis vulnerable to data. A theory that can only explain one dataset after the fact is not ready; a theory with predicted stress tests can be investigated.

2. The PLAMB Conjecture

The current PLAMB working conjecture has two active strands.

The **first strand** is a redshift-distance strand. Peter Lamb's latest diagnostic begins with the observed distance modulus rather than a model-dependent cosmological distance.

It converts distance modulus directly to an empirical luminosity-distance proxy, applies a simple first-order time-projection factor, and then tests linearity against redshift:

Core time-projection diagnostic

```
D_L,raw [Mpc] = 10^((MU - 25) / 5)
D_tau = D_L,raw / (1 + beta z)
```

Peter Lamb's current correction corresponds to beta = 0.5, so D_tau = D_L,raw / (1 + z/2). A through-origin fit then tests D_tau = (c/H0) z.

The **second strand** is a galaxy-acceleration strand. The observed radial acceleration relation in SPARC can be written as an interpolation between baryonic acceleration and measured centripetal acceleration.

The recent PLAMB/RAR derivation found that the exact RAR interpolation is recovered if an optical depth obeys tau = sqrt(g_bar / g0), with g0 close to c H0 / 2pi over the relevant H0 range.

This does not prove the physical origin of RAR, but it gives PLAMB a compact target: derive an action-count or transition-probability variable that naturally produces the square-root optical depth.

Core SPARC optical-depth branch

```
tau = (g_bar / (kappa c H0 / 2pi))^p
nu = 1 / (1 - exp(-tau))
```

The July 13 SPARC runs find p near 0.48-0.54 across the main splits, close to the square-root p = 0.5 branch.

A **third strand**, less mature, is a Noether/conservation strand. The code currently includes phenomenological variables for effective c drift, magnitude or inertia drift, and propagation dimming.

It now distinguishes free PLAMB fits from two diagnostic conservation modes: G/c constant and G/c^2 constant.

The audit is explicit that this is not yet an action-level theory: there is no Lagrangian, stress-energy tensor, conserved current, or full field equation.

That separation matters. Without it, PLAMB would overstate what its present implementation can prove.

3. Faster-than-Light Contexts and What They Do Not Prove

Because PLAMB touches light propagation and the meaning of c, it must be placed beside familiar cases where faster-than-light language appears.

The key distinction is between local signal propagation in a local inertial frame and coordinate, phase, group, or correlation effects.

Special relativity forbids controllable information or massive particles from locally outrunning light in vacuum.

General relativity and quantum theory contain phenomena that sound superluminal but preserve that operational causal rule.

Context	What can exceed c?	Why it does not prove PLAMB
Cosmic expansion and inflation	Proper separation of distant comoving regions can grow faster than c; during inflation this is extreme.	This is metric expansion, not an object locally moving through space faster than light. It shows that c-limit language is coordinate-sensitive in cosmology.
Superluminal recession today	Galaxies beyond the Hubble sphere can have recession velocities above c in standard GR coordinates.	Davis and Lineweaver show this does not violate special relativity. It is a GR horizon and distance-definition issue.
Quantum entanglement	Bell correlations violate local-realist inequalities across spacelike separation.	No-signalling remains intact. Entanglement correlations cannot be used to send a controllable message faster than light.
Cherenkov radiation	A charged particle can move faster than light travels in a medium.	The particle remains below vacuum c. The relevant light speed is c/n in the medium.
Anomalous dispersion and pulse group velocity	Group velocity can exceed c or become negative in engineered media.	The signal/front velocity and causal information transfer remain compatible with relativity.
Hypothetical tachyons or historical anomalies	Some theories allow tachyonic modes; OPERA-like neutrino anomalies were reported then withdrawn.	There is no confirmed elementary particle travelling above vacuum c. PLAMB cannot rely on such claims.

The relevance for PLAMB is therefore narrow but real. These examples do not make faster-than-light matter transport acceptable. They do show that cosmological distance, light propagation in media, and quantum correlation language require precision. A responsible PLAMB paper should say exactly which quantity is changing: local c, an effective path law, a clock-rate projection, a calibration convention, or a coordinate distance definition.

4. Data Sources

The recent PLAMB tests use public cosmology and astrophysics datasets together with local diagnostic products generated on July 12-14, 2026. The emphasis is on cross-probes: supernova Hubble diagrams, BAO and CMB background constraints, galaxy rotation curves, quasar anisotropy scans, and gravitational-wave distance summaries.

Dataset	Scale / contents	Role in this paper
Pantheon+ / Pantheon+SH0ES	1701 light curves of 1550 spectroscopically confirmed SNe Ia.	Main supernova tension test for raw-MU PLAMB.
DES SN5YR / DES-Dovekie	About 1600 DES likely SNe Ia plus low-redshift anchors; public light curves and covariance products.	Cleanest current support for beta near 0.5 in the raw-MU branch.
Union3 / Union3.1	Compressed SN distance products based on about 2000 cosmologically useful SNe Ia.	Lower-resolution supernova cross-check.
DESI DR2 BAO	More than 14 million galaxies and quasars over three years of DESI operation.	Updated BAO background test that weakens older SU2/SU2R claims.
Planck 2018 and ACT DR6	CMB temperature, polarization, lensing, and compressed/extended cosmology constraints.	Anchors background consistency and tests effective dark-sector branches.

Dataset	Scale / contents	Role in this paper
SPARC	175 late-type disk galaxies with Spitzer photometry and HI/H-alpha rotation curves.	Galaxy-scale acceleration and optical-depth PLAMB tests.
Quaia	All-sky Gaia-unWISE quasar catalogue with millions of quasar candidates.	Large-scale anisotropy and SU2 control diagnostics.
GWTC-5.0 / O4b	Cumulative LVK catalogue totaling 390 events; O4b adds the newest public candidates.	Independent but currently summary-level distance-redshift diagnostic.

5. Methods

5.1 Supernova Raw-MU Time Projection

The raw-MU analysis avoids starting from a pre-selected cosmological distance law. It converts observed distance modulus to D_L,raw and fits a model D_obs = (c/H0) z (1 + beta z) in magnitude space. The beta = 0.5 case is Peter Lamb's proposed first-order correction.

The full covariance upgrade uses Pantheon+ covariance matrices, DES inverse-covariance products, and Union compressed-node matrices where available. A profiled-offset check separates shape information from absolute-magnitude calibration.

The July 14 joint rerun combines DES-Dovekie raw, Pantheon+SH0ES, and Union3.1 as independent covariance blocks and compares no offset, one shared offset, dataset offsets, and dataset-plus-IDSURVEY offsets.

5.2 Redshift-Convention and Local-Flow Audits

Because any nonstandard redshift law is vulnerable to local-flow and frame choices, the log-kernel branch was tested in zHEL, zHD, and zCMB forms, with low-z cuts and survey-block leave-one-out diagnostics. The goal was to detect whether a PLAMB preference survives the same frame choices that standard supernova analyses treat as systematics. The July 14 rerun also treats the zHEL-to-zHD transition multiplicatively, using (1+z) = (1+zHEL) [(1+zHD)/(1+zHEL)]^alpha rather than only an additive redshift blend.

5.3 SU2/SU2R Background Tests

The SU2/SU2R branch is currently an effective dark-sector background model, not a solved SU(2) gauge-field cosmology. It is fit to supernova, BAO, and Planck-style compressed constraints and compared with Lambda-CDM using chi2, AIC, and BIC. The July update separates older BAO-covariance results from the newer DESI DR2 BAO stack, because the evidence changes substantially.

5.4 SPARC Optical-Depth Tests

The SPARC branch compares RAR to PLAMB-inspired optical-depth and screened-power forms. The July 13 stability rerun refits the optical-depth branch by split, removes high-leverage galaxies, and checks whether the preference survives. The July 14 upgrade then promotes the branch to a hierarchical MAP model with galaxy-level distance shifts, stellar M/L multipliers, matched RAR nuisance structure, held-out predictive scoring, null controls, split stress tests, and a tau-origin audit. Negative delta objective means the PLAMB optical-depth branch is preferred over RAR under the diagnostic objective.

The 18 July identifiability sequence first derives the transformation of the FR matter and antimatter background terms, then freezes an outcome-blind NED/2MRS environment proxy, a published Kourkchi-Tully group-membership predictor, and explicit cross-fitted promotion gates. All candidate and reserved galaxies are excluded from inferential outcome tables.

5.5 Negative Controls

Quaia and GWTC-5.0 are treated as controls, not proof engines. The Quaia mock comparison checks whether observed dipoles are unusual relative to available mock summaries. The GWTC-5.0 run uses only compressed redshift and luminosity-distance summaries, so it cannot replace a hierarchical gravitational-wave likelihood with selection functions and event posteriors.

6. Results

6.1 Supernova Raw-MU Results

The supernova result is a split verdict. DES-Dovekie is the cleanest supportive dataset for Peter's beta = 0.5 time-projection correction. Under full covariance, DES raw zHD with beta fixed at 0.5 gives H0 = 67.256 and chi2/dof = 0.917; zHEL beta-free gives beta = 0.528 and H0 = 67.937. Union3.1 is also supportive at compressed resolution. Pantheon+ remains discrepant, preferring H0 near 71.7-72.1 with beta around 0.54-0.56 in comparable fits.

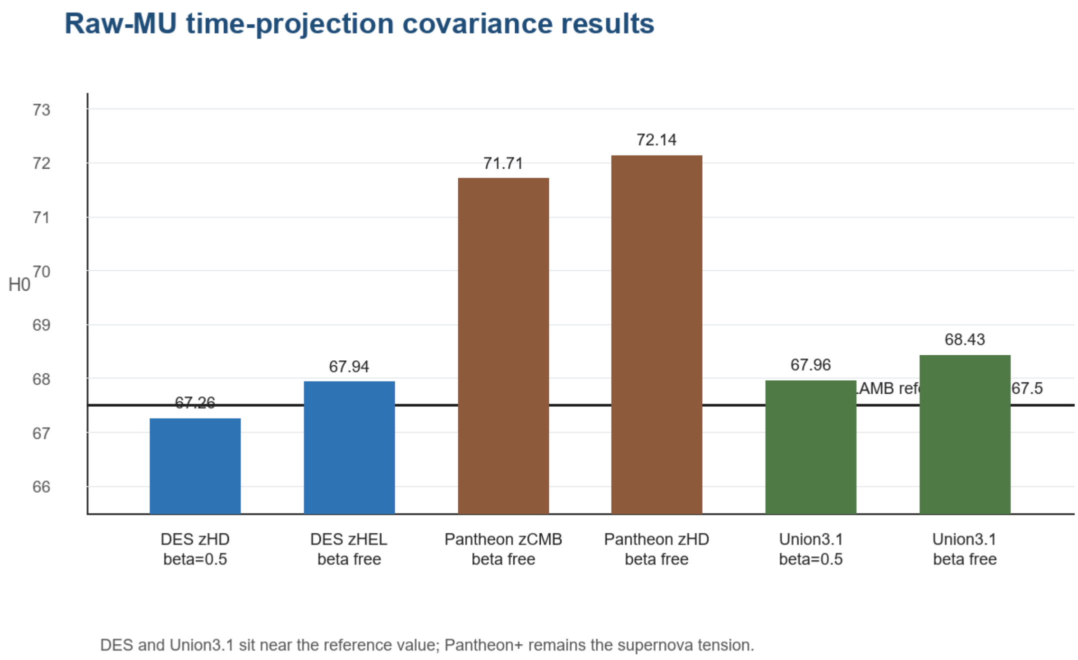


Figure 1. Selected full-covariance raw-MU time-projection results. DES and Union3.1 sit near the PLAMB reference H0, while Pantheon+ remains the main tension.

Dataset / redshift	Model row	H0	beta	chi2/dof	Readout
DES raw zHD	beta fixed	67.256	0.500	0.917	Strongly supportive of Peter correction.
DES raw zHEL	beta free	67.937	0.528	0.913	Supportive shape and H0.
DES raw zHD	beta free	68.244	0.544	0.911	Shape close to beta = 0.5.
Pantheon+ zCMB	beta free	71.709	0.542	noted as discrepant	Main supernova tension.
Pantheon+ zHD	beta free	72.140	0.555	noted as discrepant	Main supernova tension.
Union3.1	beta fixed	67.962	0.500	compressed	Supportive but low-resolution.
Union3.1	beta free	68.428	0.516	compressed	Supportive shape.

The local-flow and log-kernel audits prevent overstatement. Exact log-kernel fits can beat Lambda-CDM in zHEL for DES and Pantheon+, but the preference weakens or reverses under zHD and higher low-z cuts. By z_min = 0.05, the zHEL log preference disappears in both main SN samples. This points toward calibration, local-flow, or redshift-frame sensitivity, not an immediate cosmological replacement.

6.2 SU2/SU2R and DESI DR2

The SU2/SU2R branch changed materially after the DESI DR2 update. Older Pantheon+ plus BAO-covariance plus Planck-style fits showed a strong improvement over Lambda-CDM: SU2R had delta chi2 about -36.45, delta AIC about -32.45, and delta BIC about -21.57. With the updated DESI DR2 BAO stack, the combined SN+BAO+Planck result improves chi2 by only about 3.16, leaving AIC approximately tied and BIC favoring Lambda-CDM. The responsible conclusion is that SU2R remains useful bookkeeping, not a validated microphysical field fraction.

Run	Model comparison	Delta chi2	Delta AIC	Delta BIC	Interpretation
Older BAO-cov combined	SU2R minus LCDM	-36.451	-32.451	-21.573	Strong older preference.
DESI DR2 combined	SU2R minus LCDM	-3.163	+0.837	+11.715	Weak chi2 gain; BIC favours LCDM.
DESI DR2 BAO-only	SU2R minus LCDM	-1.217	+2.783	+3.913	Extra parameters not justified.
ACT DR6-lite proxy	SU2R PPF proxy	large penalty	not promoted	not promoted	Simple effective-w proxy performs badly.

6.3 SPARC Optical-Depth Results

The newest and most coherent positive galaxy result is the July 13 optical-depth stability rerun. Earlier fixed-exponent SPARC tests were sensitive to distance method and high-leverage galaxies; the optical-depth kappa,p branch is more stable. Across all Q2, Hubble-flow Q2, and non-Hubble Q2 splits, the exponent remains close to p = 0.5. The Hubble-flow split is the strongest: baseline delta objective = -42.872, top-5 removal = -75.561, and top-10 removal = -32.687.

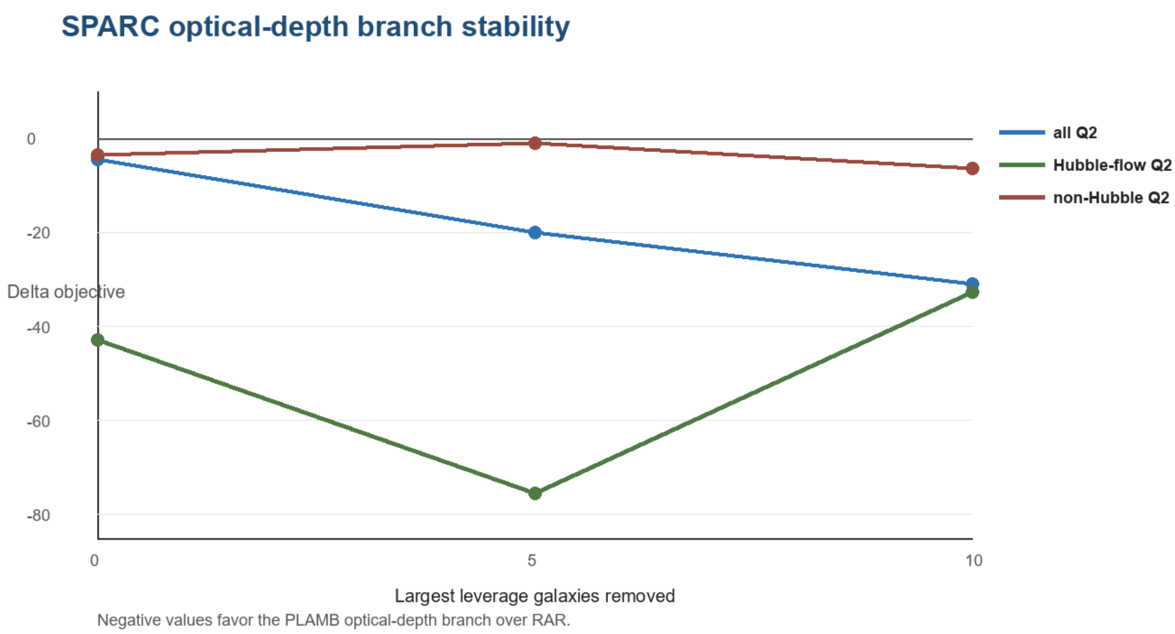


Figure 2. SPARC optical-depth stability. Negative values favour the PLAMB optical-depth branch over RAR in the diagnostic objective.

Split	N galaxies	Baseline delta	kappa	p	Top-5 delta	Top-10 delta	Readout
all Q2	163	-4.401	0.8261	0.5103	-19.987	-31.060	Survives leverage removal.
Hubble-flow Q2	92	-42.872	0.6727	0.5217	-75.561	-32.687	Strongest current SPARC result.

Split	N galaxies	Baseline delta	kappa	p	Top-5 delta	Top-10 delta	Readout
non-Hubble Q2	71	-3.480	1.0301	0.4831	-1.056	-6.427	Marginal and sensitive.

The theoretical significance is the scale.

The RAR-like acceleration g0 = c H0 / 2pi is numerically close to the fitted RAR acceleration scale for current H0 values.

At H0 = 67.5 km/s/Mpc, the local diagnostic gives g0 = 1.04374e-10 m/s^2.

This is not a derivation yet, but it provides a concrete bridge between cosmological scale-setting and galaxy rotation phenomenology.

6.4 Quaia and GWTC-5.0 Controls

Quaia and GWTC diagnostics are useful mainly because they restrain the narrative.

Quaia full-sample dipoles can look large against a tiny 50-row random-mock summary, but not against the larger 500-row zdipole mock product.

The top redshift windows therefore require full mock catalogues before they can be read as evidence for SU2 physics.

The July 15-16 object-level SU2/Quaia follow-up sharpened that statement. After regressing redshift against seven Gaia scan-law and catalogue-colour controls, the z1p00_1p50 residual family reached residual SNR = 5.7536 and global object-shuffle p = 0.000999 at |b|>10. That is a real targeted follow-up trigger, but not a promotion result: a 50,000-draw shared spatial-block wild bootstrap gives targeted-family p = 0.01352, 0.09230, and 0.23260 at 8, 12, and 16 degree cell scales.

The current Quaia claim boundary is therefore: targeted residual candidate only; no SU2 detection, no headline anisotropy claim, and no PLAMB/SU2 support unless an independent catalogue reproduces the direction and amplitude under a preregistered spatial null.

GWTC-5.0 O4b is similarly useful but not decisive: a summary-level GW-only fit prefers Lambda-CDM by AIC, while SU2/SU2R reduce chi2 slightly but are penalized for extra parameters.

Control	Recent data	Result	Consequence
Quaia	Gaia-unWISE quasar catalogue	Observed full-sample amplitudes are not unusual against larger zdipole mock summaries.	Not robust SU2 evidence yet.
GWTC-5.0 O4b	104 usable rows in local summary diagnostic	GW-only AIC winner is LCDM; SU2/SU2R have slightly lower chi2 but worse AIC/BIC.	Needs hierarchical GW likelihood.
Noether audit	PLAMB code constants and conservation modes	No action, stress-energy tensor, or conserved current yet.	Keep conservation claims as hypotheses.

6.5 July 14 Joint Supernova and Redshift-Frame Tests

Two requested follow-up tests were run on July 14. The first joint raw-MU likelihood combined DES-Dovekie raw, Pantheon+SH0ES, and Union3.1 compressed nodes using their available full covariance or inverse-covariance products as independent blocks.

It compared no offset, one shared magnitude offset, dataset-level offsets, and dataset-plus-IDSURVEY offsets.

The BIC-best rows use dataset-level offsets, not a single shared offset and not the most offset-heavy survey model.

Frame	Offset mode	Model	beta	chi2/dof	BIC and readout
HD	dataset	H0=67.5, beta free	0.5423	0.915	Best HD BIC row; supports a beta-near-0.5 shape after dataset calibration.
HD	dataset+IDSURVEY	H0=67.5, beta free	0.5113	0.907	Lower chi2, but BIC worsens because 26 parameters are counted.
HD	global	H0=67.5, beta free	0.6243	1.001	One shared offset does not recover beta near 0.5.
HEL	dataset	H0=67.5, beta free	0.5267	0.924	Best HEL BIC row; beta is closer to 0.5 than in the shared-offset row.
HEL	dataset+IDSURVEY	H0=67.5, beta free	0.5119	0.919	Again closer to 0.5 but penalized by BIC.
HEL	global	H0=67.5, beta free	0.6114	1.010	Shared-offset treatment leaves the shape too steep.

The second test reran the local-flow and redshift-frame audit with a multiplicative zHEL-to-zHD blend.

The result sharpened the main caution: the exact-log preference is strongest in zHEL-like frames and weakens or flips as the redshift convention approaches zHD.

At z_min = 0.01, Pantheon+ zHD and DES zHD favour LCDM, while zHEL favours the exact-log branch.

The multiplicative blend crosses from exact-log to LCDM between alpha = 0.45 and 0.50 for Pantheon+, and between alpha = 0.75 and 0.80 for DES.

Dataset / frame	Test point	Delta AIC log-LCDM	Preferred	Readout
DES zHD	z_min = 0.01	+1.483	LCDM	Corrected DES redshift weakens exact-log support.
DES zHEL	z_min = 0.01	-5.754	Exact log	Heliocentric DES redshift strongly favours the log branch.
DES blend	alpha 0.75 to 0.80	crosses 0	Transition	Multiplicative correction path flips the preference near the zHD end.
Pantheon+ zCMB	z_min = 0.01	-0.638	Exact log	Weak exact-log preference.
Pantheon+ zHD	z_min = 0.01	+3.202	LCDM	Corrected Hubble-diagram redshift favours LCDM.
Pantheon+ zHEL	z_min = 0.01	-3.387	Exact log	Heliocentric frame favours exact-log.
Pantheon+ blend	alpha 0.45 to 0.50	crosses 0	Transition	The preference changes near the middle of the multiplicative correction path.

6.6 July 14 SPARC Hierarchical, Null-Control, and Tau-Origin Tests

The newest SPARC work moved beyond the July 13 stability diagnostic.

A hierarchical MAP model now assigns each galaxy a distance shift with the published SPARC distance-error prior and a 3 percent floor, one stellar M/L multiplier with sigma_ln_ml = 0.25, and global disk and bulge M/L scales.

RAR was refit under the same nuisance structure. In this stricter setting, the fixed-scale tau = sqrt(g_bar/g0) branch does not beat RAR, but the kappa,p optical-depth branch remains competitive and keeps p close to one half.

plit	MAP kappa,p delta objective	p	Conservative predictive delta	Control and stress readout
all Q2	-5.386	0.50 992	+88.014 MAP-global; null baseline -10.014	Q1 only +19.933, but low-acceleration outer points -19.433. Full-sample advantage is not robust.
Hubble-flow Q2	-42.872	0.52 173	+34.680 MAP-global; null baseline +44.489	Hubble-prior blinding stays positive at +117.932, but scrambled nuisance priors flip to -45.304.
non-Hubble Q2	-3.480	0.48 306	+2.546 MAP-global; null baseline +0.753	Essentially tied in conservative scoring; scrambled nuisance priors give +33.409.

The nuisance-pull audit also matters. Pulls above 3 sigma occur in both RAR and PLAMB fits: for kappa,p versus RAR, distance-pull counts are 14/163 versus 15/163 in all Q2, 7/92 versus 6/92 in Hubble-flow Q2, and 8/71 versus 9/71 in non-Hubble Q2.

Stellar M/L pull counts are 18/163 versus 18/163, 12/92 versus 12/92, and 7/71 versus 7/71. This prevents a simple victory claim: the hierarchy is a diagnostic advance, not yet a final Bayesian evidence result.

The held-out predictive validation is also deliberately conservative.

The cleanest baseline fixes global parameters at their MAP values and marginalises held-out galaxy nuisance priors with 2048 draws rather than optimising a held-out galaxy on its own data.

Under that scorer, the kappa,p branch survives, but the advantage is concentrated: all Q2 and Hubble-flow stay positive, non-Hubble is only barely positive, and the low-acceleration outer-point stress subset favours RAR.

The tau-origin audit sharpened the theory target. With g0 = c H0 / 2pi, N = g_bar/g0 can be written as 2pi(Delta v/c) over a Hubble time, making it count-like or phase-like but not yet an action S/hbar.

If independent transition amplitudes add with random phases, tau proportional to sqrt(N) is natural; with P = 1 - exp(-tau) and g_obs = g_bar/P, the exact RAR interpolation follows.

The missing piece is a PLAMB action or conserved current that derives N, the square-root rule, and the normalisation.

Candidate origin	Current verdict	What would advance it
Transition probability	Conditional pass: P = 1 - exp(-tau) with tau = sqrt(g_bar/g0) exactly recovers the usual RAR interpolation.	Derive tau independently rather than inserting it as an ansatz.
Action-count random walk	Promising but unproven: N = g_bar/g0 is dimensionless and can act like a transition-count variable.	Define the elementary PLAMB transition and show why RMS amplitude gives sqrt(N).
Noether horizon scale	Scale pass, shape incomplete: g0 = c H0 / 2pi is natural as a horizon clock scale but does not force the bridge law.	Write an action, symmetry, conserved current, and flux or phase rule that produces N.
Current empirical stress	Important caution: the low-acceleration outer-point subset currently favours RAR by -19.433 in total log score.	Require the overnight posterior sampler to preserve p near 0.5 and survive low-acceleration predictive scoring.

The blocked overnight posterior sampler and post-run analyzer have been set up and smoke-tested.

The first completed all-Q2 chain-based analysis found that the full-sample PLAMB kappa,p branch can beat RAR in aggregate, but the result is not uniform by galaxy family or stress subset.

This made the outlier and filtered-family tests below necessary rather than optional.

6.7 SPARC Filtered-Family Refit Check

The posterior outlier audit identified a persistent negative high-leverage family: UGC06787, UGC03580, UGC02487, NGC5985, UGC09133, and NGC2903.

A subtractive leave-family-out score, using the already-fit July 14 posterior, looked very favourable to PLAMB: removing this family changed the all-Q2 outside score to +1552.8 log predictive units versus RAR.

That result was too optimistic to treat as final because it did not refit the global posterior.

Filtered all-Q2 check	Result	Readout
Filtered data	157 galaxies, 3001 Q2 rotation-curve points	Six high-leverage negative systems and 270 rotation-curve points removed.
MAP refit	PLAMB kappa,p delta objective = -17.984; data chi2 delta = -22.612	The filtered MAP still modestly prefers the exponent-free optical-depth branch over RAR.

Filtered all-Q2 check	Result	Readout
Full filtered posterior	p = 0.5193 [0.5144, 0.5237]; kappa = 0.7898 [0.7686, 0.8173]	The near-square-root bridge persists, but p = 0.5 and kappa = 1 are outside the 68 percent intervals.
Robust chain diagnostics	PLAMB max rank-split Rhat = 1.3082; RAR max rank-split Rhat = 1.0616	Neither passes the intended <=1.05 gate; PLAMB mixing is a material failure.
Historical prior-redrawn fit check	Delta log score = -368.314; 80 galaxies better, 77 worse	Superseded as a posterior-predictive label; nuisance parameters were drawn from priors, not paired posterior samples.

This is the most important update to the galaxy-scale interpretation.

The filtered refit does not confirm the simple leave-family-out optimism.

Once the six negative systems are removed and the global posterior is allowed to move, stress redistributes to other high-leverage galaxies, especially NGC2841.

The full 24k-sweep filtered posterior confirms the near-half exponent but not a full-sample predictive win.

Nuisance-pull counts are also nearly matched between PLAMB and RAR: distance pull >3 occurs in 13 PLAMB galaxies versus 14 RAR galaxies, and stellar M/L pull >3 occurs in 14 galaxies for both models.

Historical stress case	Prior-redrawn delta vs RAR	Better / worse	16 July status
all-Q2 baseline	-368.314	80 / 77	Historical computation only; not posterior-predictive. Previous readout: PLAMB loses on total score after full posterior refitting.
bulge removed	+99.583	83 / 55	Historical computation only; not posterior-predictive. Previous readout: PLAMB remains positive when bulge-heavy systems are removed.
gas dominated	+93.318	21 / 10	Historical computation only; not posterior-predictive. Previous readout: PLAMB positive in this smaller subset.
high-quality Q1	+274.392	51 / 44	Historical computation only; not posterior-predictive. Previous readout: PLAMB positive in the cleanest quality cut.
low-acceleration outer points	-173.603	81 / 74	Historical computation only; not posterior-predictive. Previous readout: The key low-acceleration subset still favours RAR on total score.
high inclination	-612.830	45 / 38	Historical computation only; not posterior-predictive. Previous readout: Strong negative stress case.
low inclination	-294.276	28 / 46	Historical computation only; not posterior-predictive. Previous readout: Negative stress case.

Best current statement: PLAMB's kappa,p optical-depth bridge remains structurally interesting, and p remains close to one half after the filtered-family refit. The 16 July audit does not support a full-sample win: PLAMB has max rank-normalised split Rhat = 1.3082 and minimum bulk ESS = 8.11, while paired posterior fit-density checks are in-sample rather than predictive.

The completed full posterior therefore strengthens the empirical p near one half pattern while weakening the case for promoting the filtered all-Q2 SPARC result as a robust model win over RAR.

6.8 Hierarchical Raw-MU Offsets and Pre-Registered External-Flow Audit

The July 14 joint raw-MU likelihood has now been promoted from a free or profiled offset diagnostic to an explicit hierarchical calibration model.

The model keeps the same independent block precision treatment for DES-Dovekie raw, Pantheon+SH0ES, and Union3.1 compressed nodes, but places zero-mean Gaussian priors on dataset and survey calibration offsets.

The residual model is MU_obs - MU_model = X_dataset delta_dataset + X_survey delta_survey + noise, with delta_dataset and delta_survey marginalised or profiled under their stated prior widths.

Prior config	Dataset sigma	Survey sigma	Purpose
tight	0.02 mag	0.01 mag	Tests whether only very small calibration excursions are enough.
moderate	0.05 mag	0.02 mag	Main calibration-prior scale used for the headline readout.
loose	0.10 mag	0.05 mag	Checks whether the signal depends on permissive calibration structure.
very loose	0.20 mag	0.10 mag	Stress case; useful diagnostically but too permissive for promotion.

Frame	Random offsets	Prior	Model	beta	Marginal BIC-like score	Readout
HD	dataset+survey	moderate	H0=67.5, beta=0.5	0.500000	3147.056	Best hierarchical row; calibration priors absorb most tension.
HD	dataset+survey	moderate	H0=67.5, beta free	0.521314	3151.038	Beta stays close to one half under explicit offset priors.
HD	dataset	loose	H0=67.5, beta free	0.541273	3158.339	Dataset-only offsets also support beta near one half.
HD	none	none	H0=67.5, beta free	0.466532	4118.004	No-offset joint fit remains poor.

This promoted likelihood is stronger statistical hygiene but also a stronger caution. It no longer counts all calibration shifts as free parameters in the same way as the previous offset BIC table; instead it uses the Gaussian random-effect objective plus an Occam/log-determinant term.

Under that treatment, the raw-MU shape remains interesting, but the interpretation is explicitly calibration-mediated. The largest moderate-prior pulls still point at Pantheon+SH0ES as the cross-dataset calibration pressure point.

The redshift-frame audit was also repeated under a pre-registered external-flow convention.

The registered primary frame is the release-provided Hubble-diagram redshift zHD for Pantheon+SH0ES and DES-Dovekie, with zHEL and zCMB retained as controls where available.

Union3.1 compressed nodes expose a single z coordinate and are not eligible for object-level frame switching.

Pantheon+ provides VPEC, so the audit checked the relation VPEC = c*((1+zCMB)/(1+zHD)-1); the reconstructed VPEC has correlation 0.999951 with the release column and median absolute reconstruction error 0.788 km/s.

Dataset	Primary frame	Controls	z_min=0.01 result	Readout
DES-Dovekie raw	zHD	zHEL	zHD delta AIC = +1.483; zHEL delta AIC = -5.754	Registered external-flow frame favours LCDM; heliocentric control favours exact log.
Pantheon+SH0ES	zHD	zHEL, zCMB	zHD delta AIC = +3.202; zHEL = -3.387; zCMB = -0.638	Registered external-flow frame favours LCDM; controls recover exact-log preference.
Union3.1 UNITY1.8	z	none	Compressed nodes only	Included in joint raw-MU likelihood, not object-level frame audited.

Dataset	Multiplicative alpha crossing	Interpretation
Pantheon+SH0ES	alpha = 0.473871	The exact-log advantage disappears before the correction path reaches the full zHD frame.
DES-Dovekie raw	alpha = 0.777855	DES stays exact-log longer along the zHEL-to-zHD path, but still crosses before alpha = 1.

The resulting supernova conclusion is therefore narrower than the earlier raw-MU enthusiasm.

The half-order beta shape is still worth tracking, especially in DES-Dovekie and Union3.1, but the exact-log branch is not frame robust under the documented external-flow convention.

For promotion, PLAMB would need to survive the registered zHD treatment rather than only the heliocentric or partial-correction controls.

7. Recent July 2026 PLAMB Analysis

The last few days of work have materially clarified which parts of PLAMB are promising and which are fragile.

The following chronology summarises the local analysis products generated on July 12-14, 2026.

Date	Analysis	Main result	Status
July 12	Raw-MU multidataset diagnostic	DES favours beta near 0.5 and H0 near 67.5; Pantheon+ tensions remain; Union3.1 supportive.	Promising, dataset-tension audit needed.
July 12	Full covariance raw-MU likelihood	DES survives covariance stress test; profiled-offset beta stable.	Strongest SN support.
July 12	Log-kernel systematics audit	zHEL preference sensitive to low-z cuts and redshift convention.	Do not promote yet.
July 12	SU2 current-status synthesis	Older BAO-cov evidence strong; DESI DR2 weakens evidence substantially.	Keep SU2R diagnostic.
July 13	Noether/constants audit	Conservation modes implemented, but no action-level theory exists.	Good theoretical hygiene.

Date	Analysis	Main result	Status
July 13	Quaia and GWTC-5.0 controls	No decisive support from available summary controls.	Negative-control discipline.
July 13	SPARC fixed-exponent and Monte Carlo tests	Fixed-exponent PLAMB is distance-method and leverage sensitive.	Caution.
July 13	PLAMB-to-RAR derivation	Exact RAR recovered by optical-depth tau = sqrt(g_bar/g0).	Theoretical target identified.
July 13	SPARC optical-depth stability	kappa,p branch near p = 0.5 survives leverage removal, strongest in Hubble-flow Q2.	Freshest positive galaxy result.
July 14	Joint DES/Pantheon+/Union3.1 offset likelihood	Dataset-offset rows are BIC-best; beta = 0.542 in HD and 0.527 in HEL with H0 fixed.	Promising but calibration-sensitive.
July 14	Multiplicative redshift-frame audit	Exact-log preference is strongest near zHEL and flips toward zHD; Pantheon+ crosses alpha 0.45-0.50, DES 0.75-0.80.	Redshift-frame sensitivity confirmed.
July 14	Hierarchical raw-MU offset likelihood	Explicit dataset and survey Gaussian priors keep HD beta-free H0-fixed beta near 0.521 under moderate dataset+survey priors.	Completed; supports a calibration-mediated, not calibration-free, readout.
July 14	Pre-registered external-flow redshift audit	Registered zHD primary frames favour LCDM for both Pantheon+ and DES; multiplicative alpha zero-crossings are 0.474 and 0.778.	Completed; exact-log branch is frame sensitive.
July 14	SPARC hierarchical MAP	kappa,p gives p = 0.510, 0.522, and 0.483 across all-Q2, Hubble-flow Q2, and non-Hubble Q2; strict fixed-scale tau does not beat RAR.	Promising, but nuisance pulls require posterior checks.
July 14	SPARC predictive/null controls	Conservative predictive kappa,p deltas are +88.014, +34.680, and +2.546, but all-Q2/null baselines are near tied and low-acceleration outer points give -19.433.	Survives controls, not a clean win.
July 14	Tau-origin derivation audit	Transition-probability and random-walk action-count routes can reproduce RAR mathematically; no action/current derivation exists yet.	Theory target sharpened.
July 14	Posterior outlier and leave-family-out audit	UGC06787 and the persistent negative high-leverage family dominate the apparent PLAMB loss; subtractive outside-family all-Q2 score is +1552.8.	Useful diagnostic, but not a refit.
July 14	Filtered-family full posterior refit	After excluding six persistent negative systems, p = 0.5193 [0.5144, 0.5237], but all-Q2 posterior-predictive score is -368.314 versus RAR.	Completed; p pattern persists, full-sample predictive case weakens.
July 16	Robust split convergence and paired-posterior nuisance audit	RAR max rank-split Rhat = 1.0616; PLAMB = 1.3082. Paired in-sample fit-density deltas are positive but not predictive.	Claim corrected: subset wins, not a full-sample win.

7.1 Computational Programs and Outputs

The July 2026 results in this paper were generated by local Python and command-wrapper programs in the Cosmology working tree.

Unless otherwise stated, source programs are under C:/Users/clive/Documents/Cosmology and outputs are under plamb_runs/diagnostics.

The runtime used for this paper build was Python 3.12.13 from the Codex bundled workspace runtime.

The July 14 science reruns used C:/Users/clive/anaconda3x/python.exe, Python 3.13.9, because that environment has Astropy support for the Union3.1 FITS compressed-node products.

The PLAMB programs themselves are exploratory research software and should be treated as part of the analysis provenance, not as independently validated public packages.

Core PLAMB / SU2 sampler

Programs: PLamb_Test_10V6c_plus.py

Primary outputs: updated_datasets_20260710/runs; plamb_conservation_smoke

Use in paper: Defines the LCDM, PLAMB, SU2, and SU2R model branches and conservation-mode scaffolds.

Direct time-projection check

Programs: diagnose_plamb_time_projection.py

Primary outputs: plamb_time_projection

Use in paper: Generated the D_L/(1+z/2) straight-line diagnostic and DES/Pantheon headline rows.

Raw-MU multidataset and covariance likelihood

Programs: diagnose_rawmu_fr_multidataset.py; fit_rawmu_fr_likelihood.py; fit_rawmu_fr_cov_likelihood.py; run_rawmu_fr_cov_likelihood.cmd

Primary outputs: rawmu_fr_multidataset; rawmu_fr_cov_likelihood

Use in paper: Generated DES, Pantheon+, Union3, and Union3.1 beta/H0 comparisons under diagonal and full covariance treatments.

Raw-MU robustness and kernel variants

Programs: validate_plamb_rawmu_holdout.py; validate_plamb_rawmu_crossdataset.py; compare_plamb_rawmu_extended_shapes.py; compare_plamb_rawmu_exact_kernels.py; compare_plamb_rawmu_constrained_gamma.py; fit_plamb_rawmu_nuisance.py

Primary outputs: plamb_rawmu_holdout; plamb_rawmu_crossdataset; plamb_rawmu_exact_kernels; plamb_rawmu_constrained_gamma; plamb_rawmu_nuisance

Use in paper: Supported the holdout, cross-dataset, exact-kernel, constrained-gamma, and survey-offset cautions.

Redshift-frame, local-flow, and H0 calibration audits

Programs: audit_plamb_local_flow_redshift.py; monte_carlo_plamb_local_flow.py; diagnose_plamb_log_kernel_residuals.py; diagnose_pantheon_survey_id_projection.py; scan_plamb_h0_calibration.py

Primary outputs: plamb_local_flow_redshift_audit_overnight; plamb_local_flow_monte_carlo_overnight; plamb_log_kernel_systematics_audit; plamb_h0_calibration_scan_projected

Use in paper: Generated the redshift-convention, low-z-cut, local-flow, IDSURVEY, and projected-H0 caveats.

Joint raw-MU offset likelihood

Programs: run_joint_rawmu_offset_likelihood.py

Primary outputs: joint_rawmu_offset_likelihood_20260714

Use in paper: Generated the DES-Dovekie raw, Pantheon+SH0ES, and Union3.1 joint covariance-block likelihood with shared, dataset, and survey-level magnitude offsets.

Joint raw-MU hierarchical offset likelihood

Programs: run_joint_rawmu_hierarchical_offset_likelihood.py

Primary outputs: joint_rawmu_hierarchical_offset_20260714

Use in paper: Promoted the joint raw-MU offset diagnostic to Gaussian dataset and survey calibration priors, with marginal random-effect scoring.

Multiplicative redshift-frame audit

Programs: audit_plamb_local_flow_redshift.py --blend-mode multiplicative

Primary outputs: plamb_redshift_multiplicative_audit_20260714

Use in paper: Generated the July 14 zHEL-to-zHD multiplicative transition, low-z-cut, local-flow, and velocity-cut redshift-frame results.

Pre-registered external-flow redshift audit

Programs: audit_plamb_local_flow_redshift.py --blend-mode multiplicative; external_flow_preregistration_20260714.json

Primary outputs: plamb_redshift_external_flow_preregistered_20260714

Use in paper: Documented zHD as the release-provided external-flow primary frame for Pantheon+ and DES, validated Pantheon+ VPEC reconstruction, and recorded multiplicative alpha crossings.

Noether and constants audit

Programs: audit_plamb_constants_noether.py

Primary outputs: plamb_constants_noether

Use in paper: Documented which PLAMB variables are Noether-sensitive and which are merely numerical or observational constants.

SU2/SU2R status and DESI transition

Programs: summarize_su2_current_status.py; diagnose_bao_transition_su2r.py; diagnose_su2r_growth_likelihood.py; diagnose_su2r_perturbations.py

Primary outputs: su2_current_status; bao_transition_su2r; growth-likelihood diagnostics

Use in paper: Generated the older BAO-cov versus DESI DR2 comparison and the growth/ACT perturbation cautions.

Quaia quasar controls

Programs: diagnose_su2_quaia_scan.py; diagnose_su2_quaia_mock_controls.py; diagnose_su2_quaia_hht_controls.py

Primary outputs: su2_quaia_scan; su2_quaia_mock_controls; su2_quaia_hht_controls

Use in paper: Generated the quasar dipole scan and mock-control interpretation.

GWTC-5.0 / O4b diagnostic

Programs: diagnose_su2_gwtc5_o4b.py

Primary outputs: su2_gwtc5_o4b

Use in paper: Generated the summary-level gravitational-wave distance-redshift cross-check.

SPARC fixed-exponent and systematic audits

Programs: diagnose_plamb_sparc_rotation.py; diagnose_plamb_sparc_fixed_exponent_grid.py; audit_sparc_distance_method_drivers.py; jackknife_sparc_high_leverage.py; monte_carlo_sparc_distance_ml_uncertainty.py; diagnose_plamb_sparc_h0_nuisance.py

Primary outputs: sparc_rotation_audit; sparc_rotation_fixed_exponent_grid; sparc_distance_method_audit; sparc_high_leverage_jackknife; sparc_distance_ml_monte_carlo; sparc_rotation_h0_nuisance

Use in paper: Generated the distance-method, leverage, Monte Carlo, and H0-nuisance cautions for the SPARC branch.

SPARC optical-depth stability

Programs: diagnose_sparc_optical_depth_stability.py

Primary outputs: sparc_optical_depth_stability

Use in paper: Generated the kappa,p optical-depth results and Hubble-flow/non-Hubble stability table.

SPARC hierarchical MAP and predictive validation

Programs: fit_sparc_hierarchical_map.py; validate_sparc_hierarchical_posterior_predictive.py

Primary outputs: sparc_hierarchical_map/optical_depth_hierarchical_20260714; sparc_posterior_predictive_loo/posterior_predictive_20260714_map_global

Use in paper: Generated the July 14 galaxy-level distance/M-L hierarchy and conservative held-out predictive scores.

SPARC null controls and split stress tests

Programs: run_sparc_null_stress_controls.py

Primary outputs: sparc_null_stress_controls_20260714

Use in paper: Generated the nuisance-prior scrambling, distance-label shuffling, Hubble-prior blinding, and split-subset stress results.

SPARC overnight posterior sampler and analyzer

Programs: sample_sparc_hierarchical_posterior.py; analyze_sparc_posterior_run.py; run_sparc_posterior_sampler_overnight.cmd; run_sparc_posterior_analyzer.cmd

Primary outputs: sparc_hierarchical_posterior/posterior_overnight_20260714

Use in paper: Generated the all-Q2, Hubble-flow, and non-Hubble chain-based posterior summaries and posterior-predictive galaxy scores.

SPARC outlier and filtered-family refit

Programs: sparc_outlier_residual_and_lofo.py; make_sparc_filtered_family_inputs.py; sample_sparc_hierarchical_posterior.py; analyze_sparc_posterior_run.py

Primary outputs: sparc_hierarchical_map/optical_depth_minus_persistent_negative_20260714; sparc_hierarchical_posterior/posterior_minus_persistent_negative_full_20260714

Use in paper: Generated the radius-resolved NGC2841/UGC06787 residuals, subtractive leave-family-out audit, filtered MAP refit, and full filtered posterior check.

PLAMB-to-RAR derivation

Programs: diagnose_plamb_rar_derivation.py; audit_plamb_constants_noether.py

Primary outputs: plamb_rar_derivation; plamb_constants_noether

Use in paper: Generated the tau = sqrt(g_bar/g0) RAR bridge, g0 = c H0 / 2pi scale table, and Noether/action-count caution.

Report synthesis

Programs: build_plamb_research_paper.py

Primary outputs: Codex work/ and outputs/

Use in paper: Generated this DOCX, its summary figures, and the document-level synthesis of results.

8. Importance of the Results

The results matter for three reasons. First, they convert a broad philosophical claim about redshift into quantitative tests.

A beta = 0.5 time-projection rule, an exact logarithmic kernel, or an optical-depth RAR bridge can be accepted, rejected, or modified by data.

That is the difference between a conjecture and a research programme.

Second, the tests identify where the pressure points are. If DES and Union3.1 support the raw-MU correction but Pantheon+ does not, the next problem is not rhetorical.

It is calibration, covariance, absolute-magnitude handling, survey offsets, and redshift-frame convention.

The July 14 joint likelihood makes this sharper: dataset-level offsets improve the joint supernova shape, but a single shared offset does not; the promoted hierarchical model keeps beta near one half only by making calibration priors explicit.

The preregistered zHD audit then makes the redshift-frame pressure point sharper still, because zHD favours LCDM for both Pantheon+ and DES.

The July 14 SPARC work makes the galaxy branch sharper in the same way: the hierarchy and predictive controls keep the optical-depth branch alive, but they also expose near-ties outside Hubble-flow, nuisance-pull limits, and a negative low-acceleration outer-point stress result.

Third, the optical-depth bridge may connect two long-standing empirical regularities: the cosmic acceleration scale c H0 and the galaxy-scale RAR acceleration.

A successful derivation would not merely fit rotation curves; it would explain why the square-root optical-depth form appears.

That would be scientifically important even if the larger no-expansion PLAMB hypothesis fails.

Fourth, the newest controls are valuable because they are uncomfortable.

A model that survives only when metadata are visible, nuisance priors are left unscrambled, or the easiest split is selected is not ready.

The current SPARC result is promising precisely because it now has a concrete posterior sampler, clear advancement criteria, and a known failure pressure point in the low-acceleration outer rotation-curve data.

9. Limitations and Falsification Tests

The current evidence is not sufficient for a replacement cosmology. The limitations are central to the paper, not footnotes.

- Pantheon+ remains the main supernova counterexample for the simple raw-MU correction.
- The joint DES/Pantheon+/Union3.1 test supports beta near 0.5 only after dataset or survey calibration offsets; a single shared offset drives beta to about 0.61-0.62.
- The promoted hierarchical raw-MU model keeps beta close to 0.5 under moderate dataset-plus-survey priors, but this is a calibration-prior-mediated result rather than a clean no-offset discovery.
- Log-kernel preferences are sensitive to redshift convention, local-flow treatment, and low-z cuts.
- The pre-registered external-flow audit chooses zHD for Pantheon+ and DES; under that registered choice both datasets favour LCDM by AIC.
- The multiplicative redshift audit confirms that exact-log support is strongest near zHEL and weakens or flips toward zHD, with alpha crossings at 0.473871 for Pantheon+ and 0.777855 for DES.
- DESI DR2 weakens the older SU2/SU2R background result; BIC favours Lambda-CDM in the updated combined stack.
- SPARC fixed-exponent PLAMB can flip under high-leverage removal and distance/M-L perturbation.
- The SPARC hierarchical MAP result is promising but still diagnostic; both RAR and PLAMB have some galaxy-level distance and M/L pulls above 3 sigma.
- Conservative SPARC predictive scoring is not a clean PLAMB win: all-Q2 and non-Hubble controls are near tied, and low-acceleration outer points favour RAR by -19.433.
- The full filtered-family posterior refit remains provisional: p stays near one half, but robust rank-split diagnostics fail and the original stress scores are not posterior-predictive because nuisance values were redrawn from priors.
- NGC2841 remains a central falsification target after the filtered refit; stress moved rather than disappeared.
- The RAR rescue improves classic convergence, but max rank-normalised split Rhat = 1.0616 still misses the approximately <=1.05 target; PLAMB remains materially non-converged at 1.3082.
- The Noether/conservation language is not yet backed by an action-level theory.
- Quaia now provides a targeted residual follow-up candidate, but the spatial-block null blocks promotion; GWTC controls do not yet provide decisive independent support.

A useful falsification standard is therefore straightforward. PLAMB should be considered disfavored in its simple beta = 0.5 form if a harmonised DES/Pantheon/Union covariance analysis with explicit offset priors shows that beta near 0.5 survives only by absorbing calibration structure, or if zHD/zCMB redshift treatments systematically erase the signal.

The SPARC optical-depth branch should be downgraded if the p near 0.5 preference disappears in the full filtered posterior chains, if kappa acts only as a distance-calibration patch, if nuisance pulls do most of the explanatory work, or if actual-chain posterior predictive checks continue to fail on low-acceleration outer points.

The SU2/SU2R branch should not be promoted unless DESI DR2, CMB, growth, and lensing constraints support it after AIC/BIC or Bayesian evidence penalties.

10. Recommended Investigations

1. Use the completed hierarchical raw-MU offset likelihood as the default supernova diagnostic, and next replace fixed prior grids with a transparent sensitivity analysis tied to external calibration budgets.
2. Treat the completed pre-registered zHD external-flow audit as the default redshift-frame readout; keep zHEL and zCMB as controls rather than headline frames unless a separate physical reason is specified in advance.
3. Do not promote the filtered-family SPARC result as a full-sample win. Require zero-divergence hierarchical fits and preregistered galaxy-held-out prediction; prohibit intrinsic matter/antimatter labels under the charge-even likelihood; keep the five reserved galaxies sealed until a separate environment design has at least 80% joint power near the earlier effect scale and passes an empirically calibrated null.
4. Turn the tau-origin audit into a derivation attempt: define a PLAMB action or conserved current, derive N = g_bar/g0 as a transition or phase count, then show why tau = sqrt(N) follows from random-phase or RMS amplitude addition.
5. Keep SU2/SU2R background work separated from PLAMB variable-c/inertia terms until a real perturbation closure is available.
6. For Quaia, freeze the z=1.0-1.5 family, mask, statistic, and spatial-null scale before any new catalogue is opened; require an independent quasar or large-scale-structure catalogue to reproduce the direction and amplitude class before interpreting the residual anisotropy as physical.
7. Build the gravitational-wave likelihood from event posteriors and selection functions, and preregister a PLAMB null-test suite with expected signs, allowed parameters, and failure thresholds.

11. Conclusion

PLAMB is still speculative, and some parts are fragile. That is exactly why the July 2026 diagnostics are useful. They separate the conjecture into claims that can be tested: a supernova time-projection correction, a redshift-frame-sensitive log kernel, an effective SU2/SU2R background branch, a Noether/conservation scaffold, and a SPARC optical-depth bridge to the radial acceleration relation.

DESI DR2 reduces the case for SU2R as a background replacement; the Quaia residual scan has produced a targeted follow-up candidate but not a spatial-null-robust anisotropy; and the SPARC optical-depth branch remains structurally interesting without a full-sample promotion.

The July 14 reruns sharpen that conclusion. The joint DES/Pantheon+/Union3.1 raw-MU likelihood does not collapse the supernova tension into one clean number: dataset-level offsets are favoured by BIC and give beta = 0.542 in HD and beta = 0.527 in HEL, while survey-heavy offsets move beta near 0.511 but pay a large information-criterion penalty.

The promoted hierarchical version improves the statistical framing: moderate dataset-plus-survey priors give HD beta-free H0-fixed beta = 0.521, but that is explicitly a calibration random-effect result.

The preregistered external-flow audit likewise strengthens the caution around the exact-log branch: with zHD registered as the primary frame for Pantheon+ and DES, both datasets favour LCDM, and the exact-log preference appears mainly along the zHEL or partial-correction controls.

The July 14 SPARC reruns are similarly clarifying. The optical-depth branch has advanced from a stability diagnostic to a hierarchy, predictive, null-control, stress-test, and filtered-family refit stage.

The fitted exponent remains close to one half in the kappa,p hierarchy, including after removing the persistent negative high-leverage family. But the result is not a simple victory over RAR. The July 16 audit shows max rank-normalised split Rhat = 1.3082 for PLAMB and 1.0616 for RAR, and it reclassifies the earlier negative stress scores as prior-redrawn in-sample checks rather than posterior-predictive evidence. The correct claim remains: subset wins, not a full-sample win.

The 18 July charge and environment sequence further narrows the claim. Complete conjugation preserves the implemented galaxy dynamics, so the previously aberrant galaxies cannot be identified as antimatter from rotation curves alone. The local 2MRS hint fails its preregistered predictive gate, published group richness is null, and the group design is underpowered at the earlier effect scale. These are constraints on the present observable and design, not evidence for a hidden antimatter population.

The most defensible conclusion is not that PLAMB has overturned standard cosmology. It has not. The conclusion is that PLAMB has become specific enough to be dangerous to itself in the productive scientific sense: it now has clear empirical successes, clear tensions, and concrete tests that can decide which pieces survive.

12. July 16, 2026 Post-Audit Correction

The change to a newer Codex reasoning model did not modify any completed chain or stored result. It prompted an independent reproducibility audit of the recent SPARC programmes. Two methodological issues were found in the analysis layer: classic unsplit Rhat understated chain problems, and the stress-score routine sampled galaxy nuisance terms from priors instead of using their paired posterior draws.

12.1 Robust convergence

```
Rhat_robust   = max(Rhat_rank, Rhat_folded)
ESS_bulk      = approximate effective draws for rank-normalised chains
MCSE_mean     = posterior SD / sqrt(ESS_bulk)
```

Model	Worst parameter	Classic Rhat	Rank-split Rhat	Bulk ESS	Readout
RAR	log_ybul	1.03071	1.06163	76.27	Improved, gate not passed
PLAMB kappa,p	log_ydisk	1.05186	1.30819	8.11	Material non-convergence

12.2 Paired posterior fit-density checks

For each retained draw, the revised audit evaluates the stored global parameters together with the associated galaxy mass-to-light nuisance values. This corrects the pairing defect, but it remains an in-sample fit-density calculation. It is not leave-one-galaxy-out, grouped cross-validation, or an external posterior-predictive test.

Case	N galaxies	N points	Delta fit-density vs RAR	PLAMB better / worse
All-Q2 baseline	157	3001	+13.602	78 / 79
Bulge removed	138	2279	+25.082	69 / 69
Gas dominated	31	449	+4.828	20 / 11
High inclination	83	1739	+11.927	43 / 40
Low-acceleration outer	155	1401	+0.890	79 / 76
Low inclination	74	1262	+1.644	36 / 38

Interpretation: the sign reversal relative to the historical stress table shows that nuisance handling mattered. It does not promote PLAMB because the PLAMB chains fail robust convergence and the statistic reuses the fitted galaxies.

12.3 Corrected claim and next programme

12.4 SU2/Quaia residual-global update

The July 15 residual-global object-shuffle search found a strong controlled residual candidate in the z1p00_1p50 family: |b|>10 gives residual SNR = 5.7536, residual priority = 0.05524, and global p = 0.000999 after scan-law and colour regression. The |b|>15 row is nearly identical.

The July 16 spatial-block and template-collinearity audit then supplied the stricter promotion readout. Common-sign spatial wild bootstrap p-values are 0.01352, 0.09230, and 0.23260 for 8, 12, and 16 degree cells, while the locked |b|>15 direction has scan-law/colour projection R2 = 0.16363 and VIF = 1.19565. This is moderate template alignment, not catastrophic collinearity, but it is enough to keep the result at targeted follow-up status.

Corrected Quaia boundary: the candidate is worth an independent preregistered validation, but it is not a SU2 detection, not a PLAMB confirmation, and not a headline anisotropy result.

The near-half exponent remains a useful structural target and the filtered MAP/subset results remain worth following. The publication-grade next step is not simply more sweeps of the same parameterisation. Reparameterise the mass-to-light hierarchy, use a proven HMC/NUTS implementation where feasible, require robust split Rhat <= 1.01-1.05 with adequate bulk and tail ESS, and only then run preregistered galaxy-held-out or grouped cross-validation.

CORRECTED CLAIM BOUNDARY: PLAMB has subset and structural wins, not a full-sample win. Robust convergence and out-of-sample predictive superiority over RAR are both required before promotion.

13. 18 July 2026 FR Charge, Environmental Replication and Power Calibration

13.1 Complete charge conjugation and identifiability boundary

The Full Relativity (FR) source distinguishes matter and antimatter background contributions, denoted B_M and B_AM. Under complete charge conjugation, every matter contribution is exchanged with its antimatter counterpart. Defining B_+ as the total background magnitude and chi as the signed fractional asymmetry gives the aligned transformation

$$
\begin{aligned}
B\_M' &= B\_AM \\
B\_AM' &= B\_M \\
B\_+' &= B\_M' + B\_AM' = B\_+ \\
chi' &= -chi \\
|chi'| &= |chi|
\end{aligned}
$$

Here B_M and B_AM are the matter and antimatter background terms, B_+ = B_M + B_AM, and chi = (B_M - B_AM)/B_+. The implemented PLAMB optical-depth rotation law depends on charge-even totals or magnitudes. Complete conjugation therefore preserves its predicted galaxy dynamics. A host-only swap inside a separately fixed signed environment could be non-invariant, but FR presently supplies neither an observed signed matter-minus-antimatter map nor a fully specified inertia response f(|chi|, B_+, v).

The preregistered 12-scenario conventional-systematics matrix consequently found the rotation-curve matter/antimatter label non-identifiable. Three of six previously selected galaxies retain model-relative PLAMB tension after deterministic multi-start profiling, but the matched-block direction is incoherent. No galaxy is classified as antimatter, and the five reserved galaxies remain unopened.

13.2 Preregistered local-environment test

The first environmental test used NASA/IPAC Extragalactic Database (NED) positions and redshifts with the Two Micron All Sky Survey Redshift Survey (2MRS) as a charge-blind tracer. The 66-galaxy primary sample was fixed at SPARC quality Q <= 2, distance 10-40 Mpc and |b| >= 10 degrees, excluding all six outcome-selected candidates and all five reserved galaxies.

For galaxy i, the frozen outcome Delta_i compares the profiled PLAMB and radial acceleration relation (RAR) objectives q_P,i and q_R,i per N_i rotation points. E_i is the environmental predictor and X_i is the host-control vector:

$$
\begin{aligned}
Delta\_i &= sgn(q\_P,i - q\_R,i) \ln[1 + |q\_P,i - q\_R,i| / N\_i] \\
E\_i &= \ln N\_group,i \\
Delta\_i &= alpha + beta\_E E\_i + gamma^T X\_i + epsilon\_i
\end{aligned}
$$

The charge-blind 2MRS composite averages robust-standardised 2 Mpc and 5 Mpc neighbour counts, inverse nearest-neighbour distance and a softened K-band tidal sum. Cross-fitting controls morphology, luminosity, surface brightness, gas mass, velocity, inclination, distance uncertainty, bulge proxy, point count, distance, latitude and distance method. The primary partial Spearman association is rho = -0.307254 with two-sided permutation p = 0.0117994, but cross-validated root-mean-square error (RMSE) improves by only 2.398%. It misses the locked p <= 0.01 and >= 5% predictive-improvement thresholds and weakens in wider and stricter selections. The gate fails.

A separately labelled post hoc audit finds no single-galaxy driver: every leave-one-out rho remains negative, from -0.3472 to -0.2775. The 2 Mpc count and softened tidal component each have exploratory Holm-adjusted p = 0.0478, while the 5 Mpc count is null. These diagnostics motivate only a conventional local-environment nuisance hypothesis.

13.3 Published group-catalogue replication

A second preregistration matched non-reserved SPARC galaxies to the fixed Kourkchi-Tully catalogue of groups within 3500 km s^{-1}. Of 158 non-reserved $Q <= 2$ targets, 128 obtain unambiguous matches; the 65-galaxy primary sample contains 24 catalogue singles and 41 grouped galaxies, with N_{group} ranging from 1 to 61. The primary predictor $E_i = \ln N_{group}$ avoids choosing a group-mass cut from the outcome, although the catalogue still shares nearby-galaxy and K_s-band inputs with 2MRS and is not an independent raw survey.

Table 13.1. Locked environmental development and replication readout

Diagnostic	N	rho	p	CV RMSE change	Status
2MRS local composite	66	-0.3073	0.01180	+2.40%	Gate failed
Published group richness	65	+0.04768	0.7044	-5.41%	Null; gate failed
Catalogue group mass	65	+0.02456	0.8470	-3.47%	Secondary null
Grouped/single indicator	65	+0.02762	0.8229	-2.10%	Secondary null

Published group richness is null: rho = 0.04768, p = 0.7044, and adding richness worsens held-out RMSE by 5.41%. Luminosity-derived group mass and grouped/single status are also null, and the +90 degree shifted-sky control is larger in absolute association than the actual predictor. Broad catalogue group membership does not replicate the local 2MRS hint. The result is an independent-predictor replication on previously generated outcomes, not a held-out-outcome replication.

13.4 Injection-and-recovery power calibration

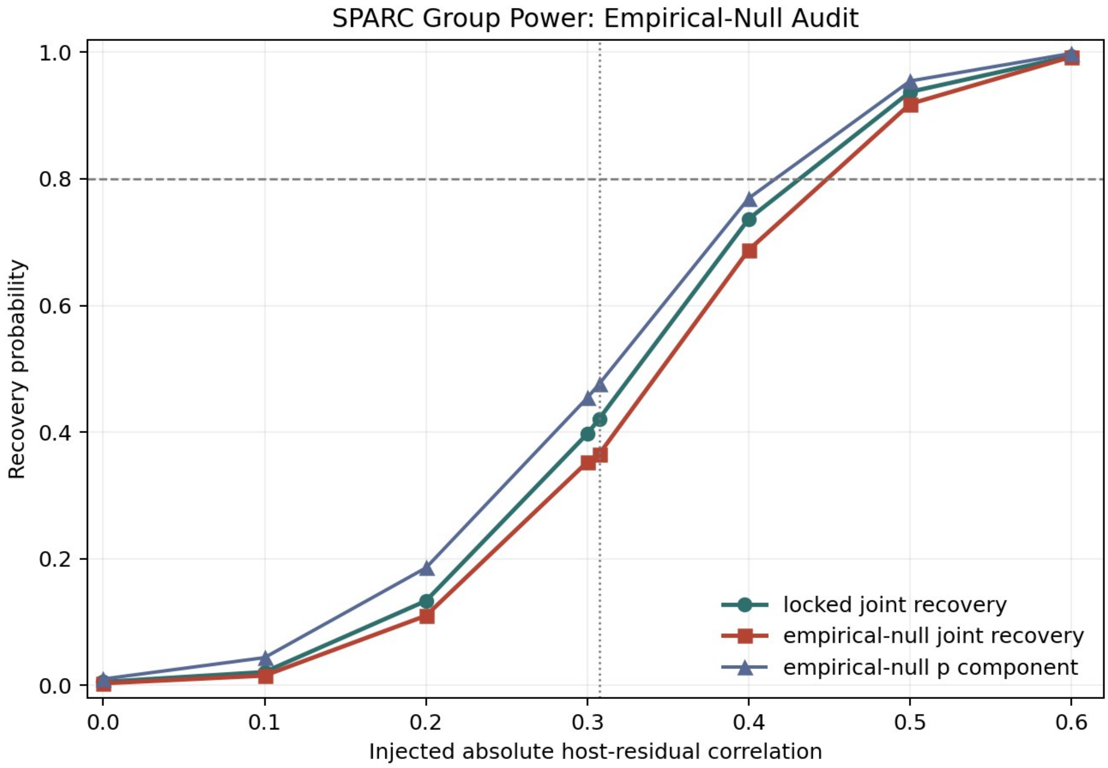
The null group result cannot be interpreted without power. A separately committed programme therefore fixes the exact 65-galaxy design, ten folds and ridge penalty before running 40,000 empirical-residual injections plus 100,000 null rank permutations. For target host-residual correlation r, replicate m is generated as

$$
\begin{aligned}
y_i^{(m)}(r) &= h_i + \sigma_e[\pi_m(e_i) + \beta(r)\,\tilde{E}_i] \\
\beta(r) &= -r / \sqrt{1 - r^2}
\end{aligned}
$$

Here h_i is the host-only ridge signal, e_i is the centred empirical residual, π_m is a without-replacement permutation, σ_e restores the observed residual scale, $\tilde{E}_i$ is host-residualised standardised group richness, and the negative sign matches the earlier 2MRS direction. Joint recovery requires $p <= 0.01$, at least 5% cross-validated RMSE improvement and a common coefficient sign in at least 8 of 10 folds.

At $|r| = 0.307254$, the locked p component recovers 66.24% and the locked joint gate 42.08%. The full empirical-noise pipeline reveals that the nominal rank-permutation p component is anti-conservative, firing in 3.26% rather than 1% of null simulations. Recalibrating the 1% threshold from $|\rho| >= 0.317526$ to $|\rho| >= 0.377185$ reduces anchor p recovery to 47.58% and joint recovery to 36.50%. The joint null rate remains conservative at 0.32% after recalibration. At least 80% joint recovery is first reached near $|r| = 0.5$.

Figure 3. Empirical-null power calibration for published group richness. The dotted vertical line marks the earlier |rho| = 0.307254 anchor and the dashed horizontal line marks 80% recovery.



13.5 Scientific implication and next gate

The July 18 sequence closes the intrinsic SPARC antimatter-classifier route under the implemented charge-even likelihood and rejects broad group richness as a replicated explanation of the earlier local-density pattern. It does not exclude an environmental effect as small as the earlier standardised association because the 65-galaxy group design has only 36.50% calibrated joint power at that scale. Conversely, it is sensitive to much larger group-richness effects near $|r| >= 0.5$.

The next defensible calculation is sample-size scaling under the empirically calibrated null, followed by a separately preregistered environment test in an independent rotation-curve and environmental data combination. The five reserved SPARC galaxies remain sealed. Unsealing requires an explicit held-out replication document and cannot be triggered automatically by development or power results.

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